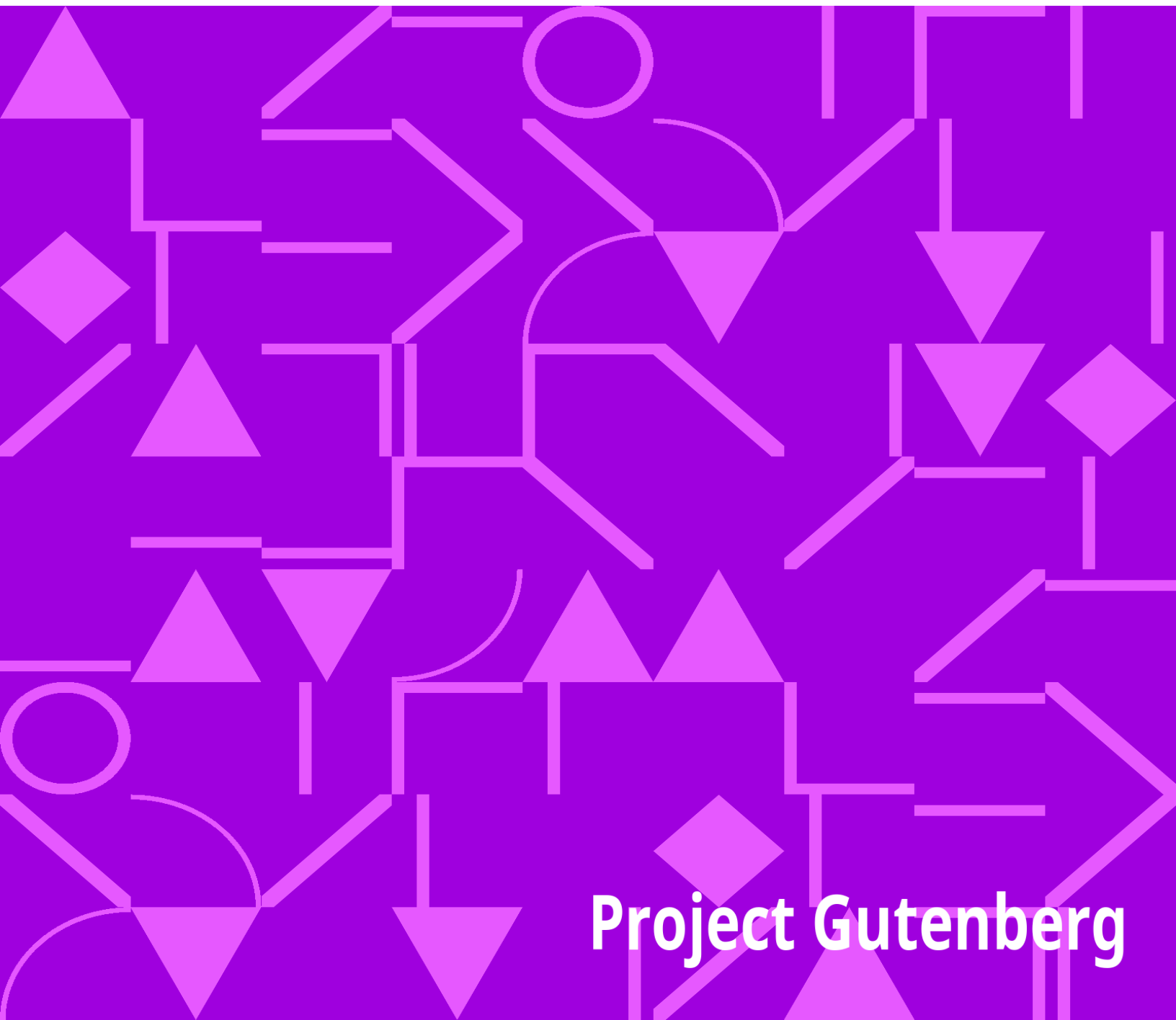


A New Bog Lemming (Genus *Synaptomys*) From Nebraska

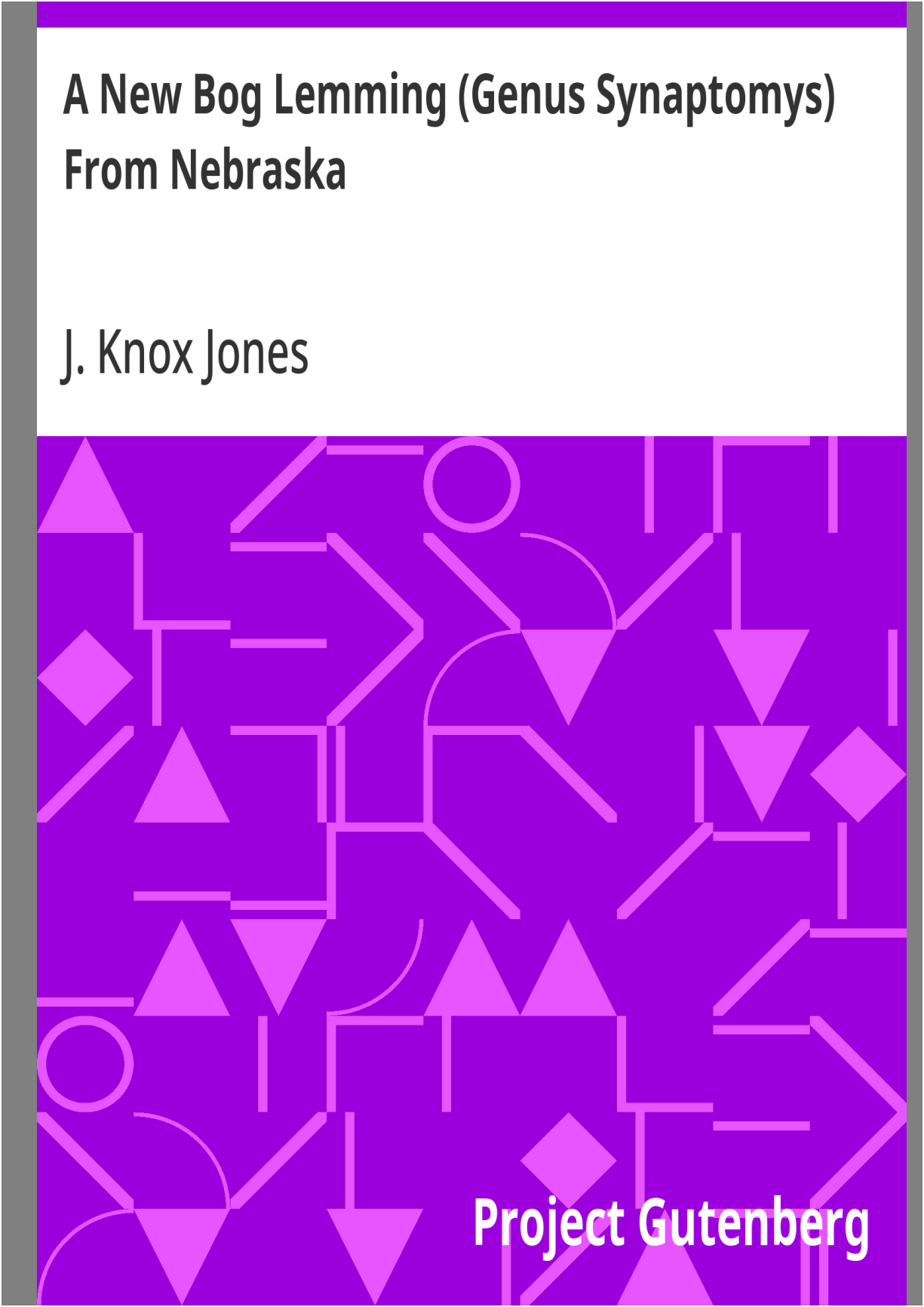
J. Knox Jones

An abstract geometric pattern composed of various shapes including triangles, squares, circles, and lines, all rendered in shades of purple and magenta. The pattern is dense and covers the entire background of the page.

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A New Bog Lemming (Genus *Synaptomys*) From Nebraska

J. Knox Jones



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LEMMING (GENUS SYNAPTOMYS) FROM NEBRASKA ***

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From Nebraska

BY

J. KNOX JONES, JR.

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A New Bog Lemming (Genus *Synaptomys*)

From Nebraska

BY

J. KNOX JONES, JR.

In the autumn of 1952, I obtained a southern bog lemming, *Synaptomys cooperi*, at Rock Creek State Fish Hatchery, Dundy County, in extreme southwestern Nebraska. This locality of record is the westernmost for the species in North America. Subsequently, I reported this specimen in the literature (Univ. Kansas Publ., Mus. Nat. Hist., 7:486, 1954), provisionally assigning it to *Synaptomys cooperi gossii*, the subspecies occurring in eastern Nebraska. In late November of 1956, J. R. Alcorn collected three additional bog lemmings at the Rock Creek Hatchery.

These specimens from Dundy County represent a relict population that differs in several characteristics from *S. c. gossii*, and that differs also from all other subspecies of the species. This relict population is, therefore, here given subspecific recognition.

***Synaptomys cooperi relictus*, new subspecies**

Type.—Adult female, skin and skull, University of Kansas Museum of Natural History no. 51617, from Rock Creek State Fish Hatchery, 5 mi. N, 2 mi. W Parks, Dundy County, Nebraska; obtained November 1, 1952, by J. Knox Jones, Jr., original no. 995.

Distribution.—Known only from the type locality.

Diagnosis.—Size large for the species, both externally and cranially; dorsal pelage dark; nasals broadly flared anteriorly (especially broad in relation to nasal length); auditory bullae small; molariform tooth-rows and incisive foramina long.

Measurements (in millimeters).—External measurements of the type specimen, followed by those of another adult female (KU 72603), are: Total length, 141, 145; length of tail-vertebrae, 24, 21; length of hind foot, 20, 20; length of ear from notch, 11, 12. The type specimen weighed 46.3 grams. Cranial measurements were taken in the manner described by Wetzel (Jour. Mamm., 36:2-3, 1955) except that he did not record the occipitonasal length. These cranial measurements of the type and KU 72603 are: Occipitonasal length, 30.2, 29.8; condylobasilar length, 27.2, 27.1; zygomatic breadth, 18.1, 17.9; lambdoidal breadth, 14.2, 13.8; length of nasals, 8.2, 8.3; breadth of nasals, 4.2, 4.0; length of rostrum, 6.6, 6.6; breadth of rostrum, 6.1, 5.9; breadth of upper incisors, 4.6, 4.2; length of maxillary tooth-row, 8.5, 8.4; length of incisive foramen, 5.8, 5.5; interorbital breadth, 3.1, 3.5.

Comparisons.—From *Synaptomys cooperi gossii* (specimens from eastern Nebraska and eastern Kansas, including one topotype), *S. c. relictus* differs in: Dorsal coloration, in comparable pelages, darker, venter lacking buffy tinge; skull averaging larger in all cranial dimensions (except rostral length, which is approximately the same), especially breadth of upper incisors and length of molariform tooth-rows; nasals broader anteriorly; auditory bullae nearly equal in size, thus relatively smaller. From *Synaptomys cooperi paludis* (holotype and paratypes) of Meade County, Kansas, *S. c. relictus* differs in: Skull averaging smaller in all cranial measurements except rostral length and breadth of upper incisors, which are approximately the same, and breadth of nasals, length of incisive foramina and length of molariform tooth-rows, which measure more; nasals relatively (48 per cent of length of nasals) as well as actually broader anteriorly; anterior border of zygomatic plate more concave; auditory bullae smaller; infraorbital foramina larger when viewed anterolaterally. *S. c. relictus* closely resembles *S. c. paludis* in color and external proportions.

Remarks.—The total population of *Synaptomys cooperi relictus* may be small because the only suitable habitat known to me for these mice is the dense, grassy area, approximately 100 yards wide and one mile long, around some of the rearing ponds and along the creek at Rock Creek Hatchery. It has been taken there in association with *Cryptotis parva parva*, *Blarina brevicauda carolinensis*, *Reithrodontomys megalotis dychei*, *Peromyscus maniculatus nebrascensis*, *Microtus ochrogaster haydenii*, and

another relic, *Microtus pennsylvanicus finitis*. All specimens of the newly named bog lemming are from the border zone between the wet-substrate habitat of *M. p. finitis* and the drier habitat occupied by *M. o. haydenii*. Approximately 3000 trap nights produced the four known specimens.

S. c. relictus, like *S. c. paludis*, represents a relict population of the more southwesterly distribution of the subgenus *Synaptomys* during Wisconsin and post-Wisconsin times. Additional relict populations likely will be found in the eastern Great Plains.

The new subspecies is intermediate in some features between *paludis* and *gossii*. The type locality is separated from that of *paludis* (14 mi. SW Meade, Meade County, Kansas) by a distance of approximately 220 miles over habitats largely unsuitable for bog lemmings. The nearest locality of record for *S. c. gossii* to the east of the type locality of *relictus* is at Hunter, Mitchell County, Kansas (see Cockrum, Univ. Kansas Publ., Mus. Nat. Hist., 7:196, 1952), approximately 200 miles distant. The locality of record of *gossii* in Nebraska nearest to the type locality of *relictus* is even farther eastward—1 mi. N Pleasant Dale, Seward County (KU 50188).

Specimens examined.—Four, from the type locality (KU 51617, 72601-03).

Transmitted March 11, 1958.

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